



Acid

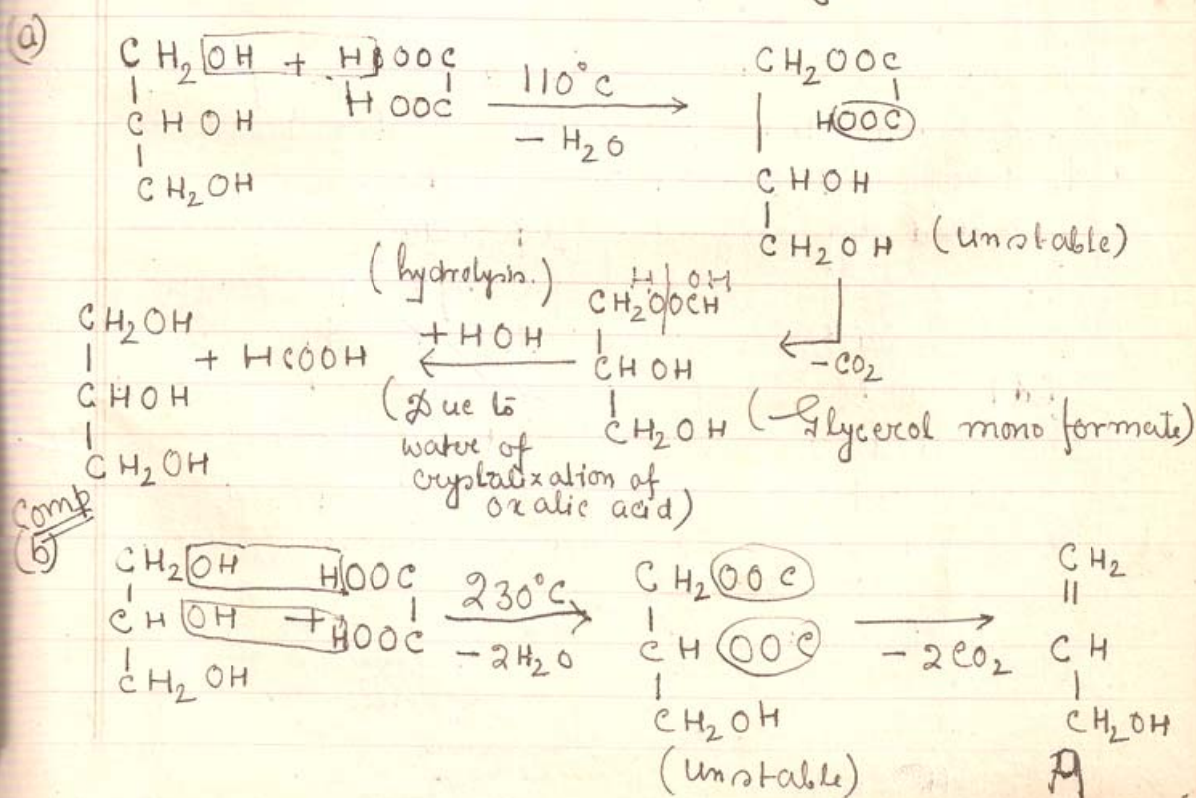
Organic acid must contain Carboxylic group ($-\text{COOH}$)

Qr. How is Formic acid prepared in the Lab. in dry and pure form. Discuss the abnormal properties of formic acid. Mention its uses.

Theory - When a mixture of oxalic acid and Glycerol is heated at 110°C we get formic acid as a distillate.

W.H.

Glycerol acts as if a catalyst.





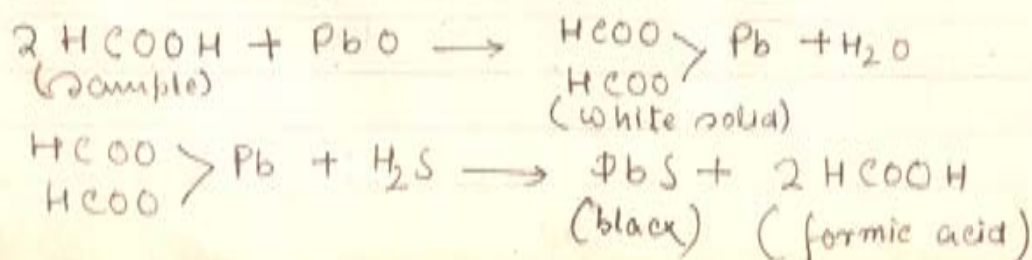
Procedure → In a distilling flask about 50 gms of Oxalic acid (solid) is taken. It is fitted with a dropping funnel through which about 50 cc of Glycerol is added into it. It is fitted with a thermometer to record the temperature 110°C . The flask is fitted with a water condenser as shown in the figure.

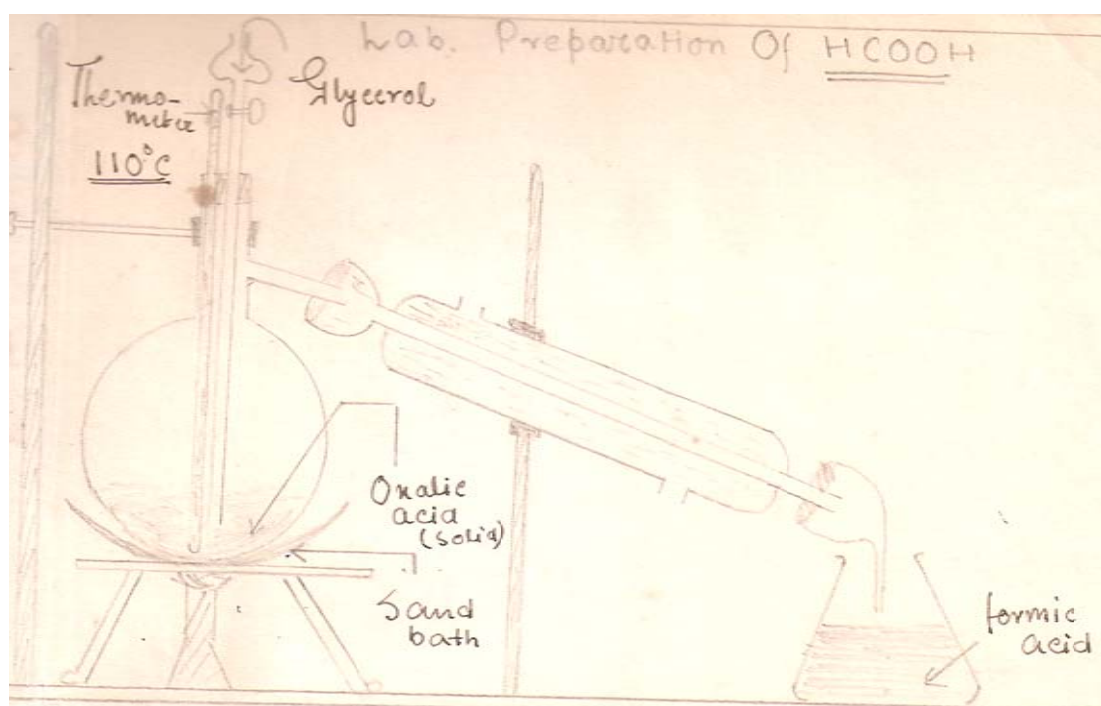
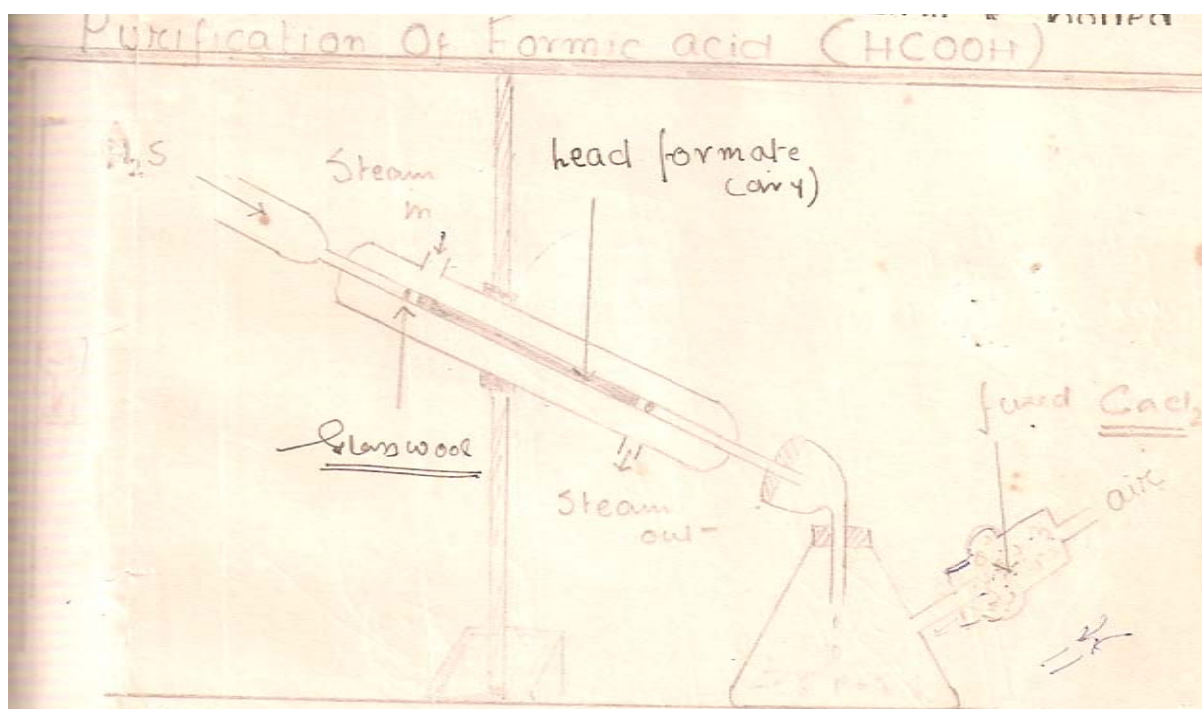
On heating the mixture in a sand bath between 110°C – 115°C , we get vapour of formic acid. On cooling the vapour by water condenser we get liquid of formic acid in the receiver.

Precaution → Sand bath should be used to get the uniform temperature as heating should be done in between 110° – 115°C . We know at higher temperature other product is formed. 230°

Purification → Formic^{acid} has the impurity only water. The boiling point of formic acid is 100.5 and that of water is 100°C so they cannot be separated by fractional distillation.

The hydrate sample of formic acid is treated with PbO to get lead formate which are separated, dried. The dried sample of lead formate is fitted in the condenser tube and decomposed by dry H_2S when formic acid (anhydrous) is obtained.

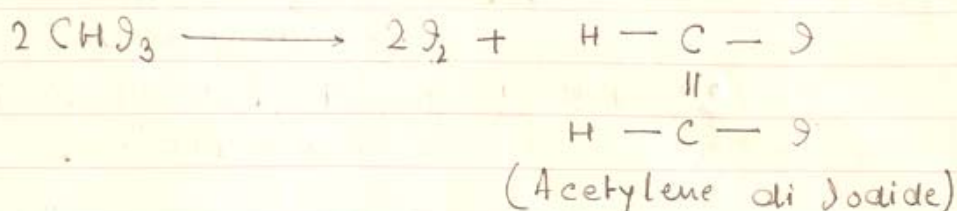






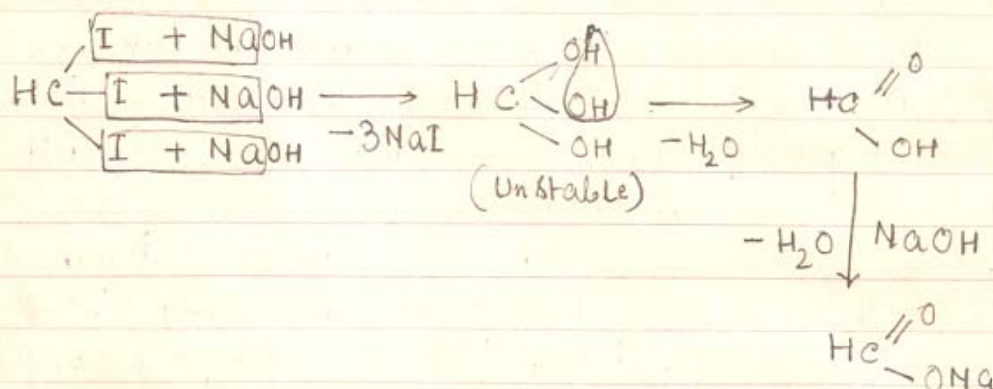
Chemical Properties

- (1) Stability :- It does not form Phosgene as does chloroform but in the presence of moist air it gives Iodine.

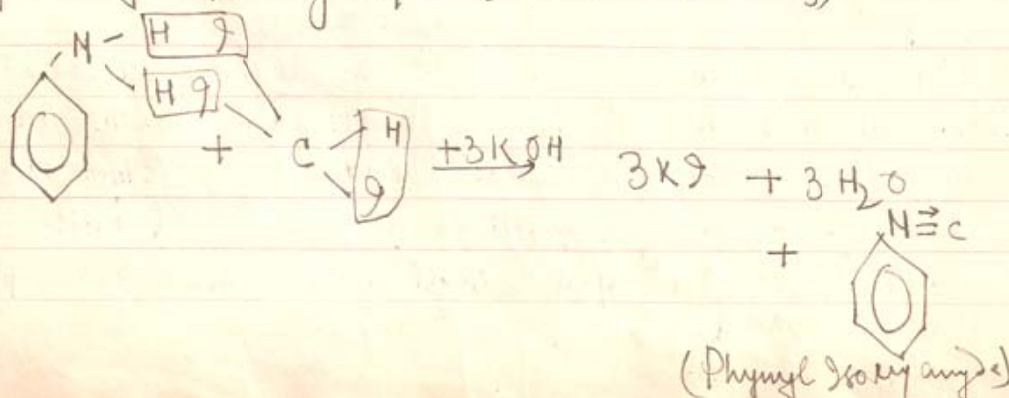


N.H

- (2) With NaOH Solution → When it is boiled with NaOH solution we get sodium formate (similar to CHCl_3)



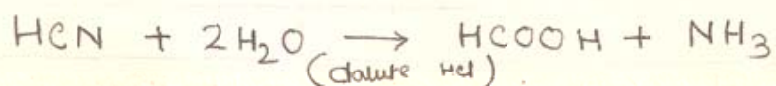
- (3) Carbylamine Reaction → When Iodoform is boiled with aniline and KOH we get a bad smell of Phenyl Iso cyanide (similar to CHCl_3)



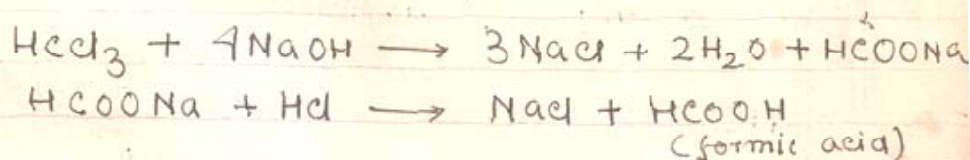


Other Methods Of Preparation

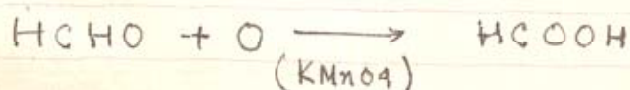
When HCN is hydrolysed with water with a little HCl (catalyst) we get formic acid.



From CHCl_3 → when Chloroform is boiled with NaOH we get Sodium formate. This Sodium formate with HCl formic acid is formed.



When formaldehyde is oxidised we get formic acid.



Physical Properties

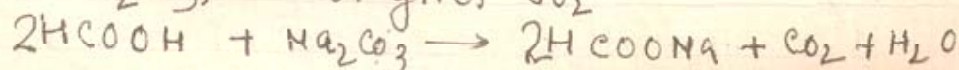
It is a colourless pungent smelling liquid. B.P 100.5°C freezes at -8.3°C . It is miscible in alcohol, ether, water. If it is poured on skin it causes blister.

Chemical Properties

With NaOH solution, — It gives sodium formate



With Na_2CO_3 , — It gives CO_2





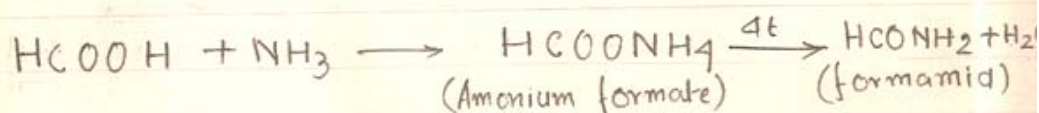
With Sodium bicarbonate; — gives CO_2



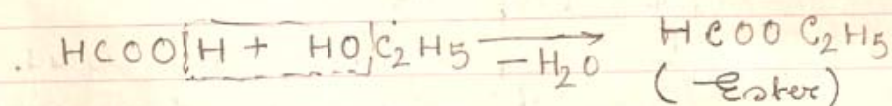
With Zn ; — it gives hydrogen.



With NH_3 — When Ammonia gas is passed through formic acid we get Ammonium formate. which on heating gives formamide (formamide)



With With alcohol — When a mixture of HCOOH and Ethyl alcohol is heated we get Ethyl formate which has fruity smell.

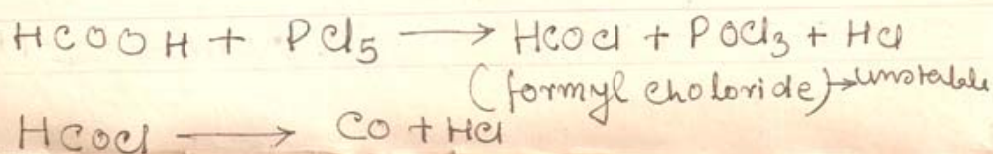


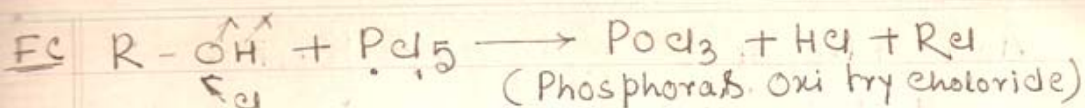
Acid + Alkali = Salt + Water.

Acid + Alcohol = Ester + Water.

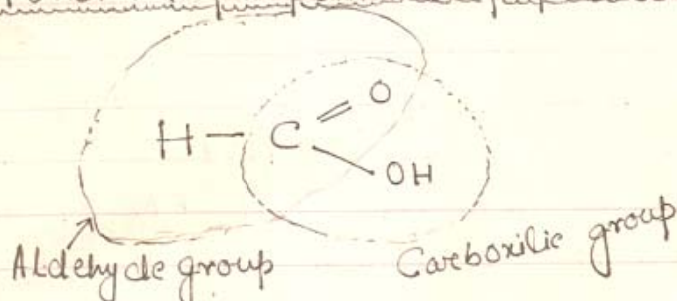
With PCl_5 — When formic acid is treated with (Phosphorus pentachloride)

PCl_5 we get formyl chloride which at once breaks into Carbon monoxide.



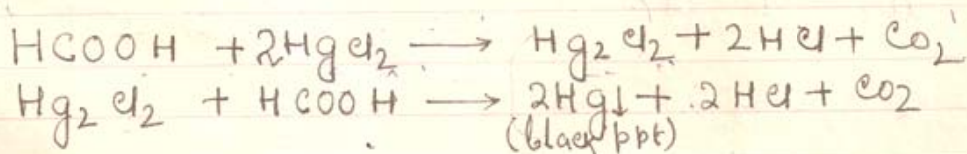


Abnormal properties of formic acid



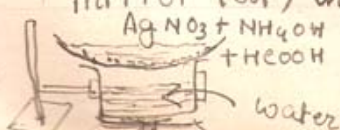
If we look at the structure of formic acid very carefully we observe that it contains not only Carboxylic group ($-\text{COOH}$) but also one Aldehyde group ($-\text{CHO}$). Therefore formic acid shows the general properties of an acid ~~but~~ also the properties of an aldehyde, this is the reason why formic acid shows the abnormal properties. It differs from acetic acid/other members of acid family.

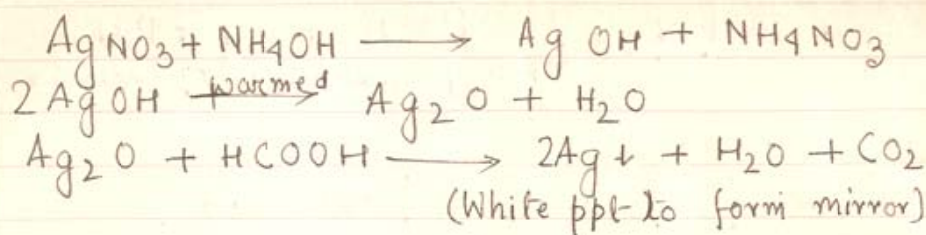
(8) With Mercuric chloride (HgCl_2) $\rightarrow$ When it is treated with HgCl_2 , at first we get Mercurous chloride Hg_2Cl_2 then it gives the ppt of Mercury.



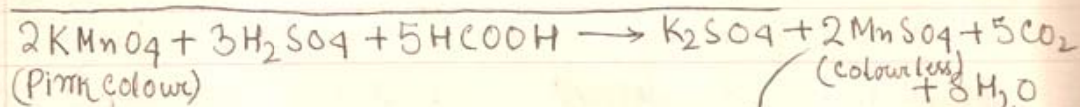
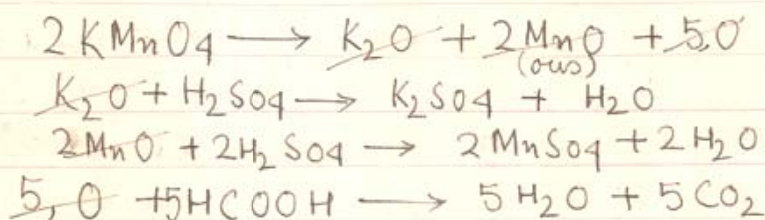
N.H
(9)

With Ammonical AgNO_3 solution ($\text{NH}_4\text{OH} + \text{AgNO}_3$) $\rightarrow$ When HCOOH is warmed with Ammonical silver nitrate solution, we get beautiful silver mirror (Silver mirror test) this is also the test of aldehyde.



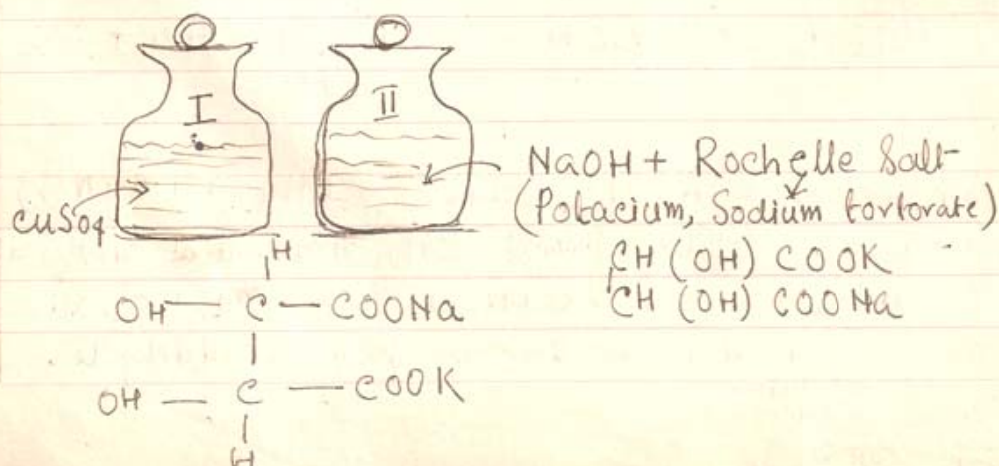


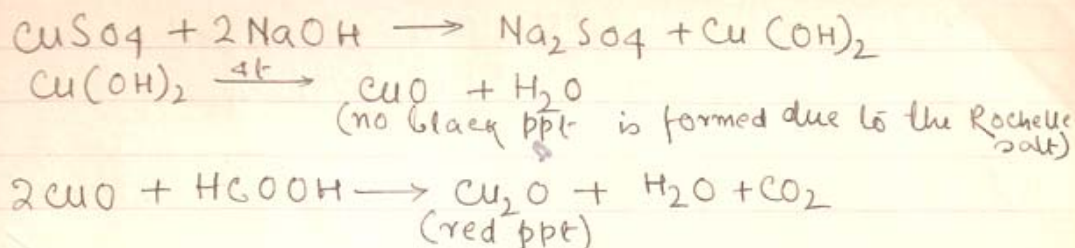
(10) With KMnO_4 Solution $\rightarrow$ When formic acid is added to KMnO_4 solution, acidified with a little dilute H_2SO_4 , the pink colour of KMnO_4 is discharged due to the formation of MnSO_4



MnSO_4 is called autocatalyst

11. With Fehling's Solution $\rightarrow$ When a mixture of Fehling's solution and formic acid is heated we get red ppt. of Cu_2O (cuprous oxide) this is also the test of $-\text{CHO}$ group





12. With Con. $\text{H}_2\text{SO}_4 \rightarrow$ It abstracts a water molecule from formic acid giving Carbon monoxide

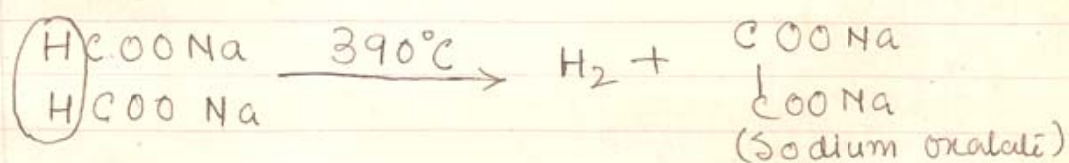


Uses of Formic acid

- (1) It is highly used in leather industry also in cotton industry.
- (2) To prepare medicine.
- (3) To preserve certain fruits for a long time.

important

R.Q. What happens when Sodium formate is heated at 390°C



Qr. In a tabular form distinguish between formic acid and acetic acid

Reagent	HCOOH	CH_3COOH
(1) With HgCl_2	White ppt of Hg	No. Such
(2) $\text{AgNO}_3 + \text{NH}_4\text{OH}$ (warmed)	Silver mirror is formed	No Such
(3) KMnO_4 solution + H_2SO_4	Pink colour of KMnO_4 is discharged	No Such



Reagent	HCOOH	CH_3COOH
(4) With Fehling Solution (boiled)	Red ppt of Cu_2O	No such
(5) With Con. H_2SO_4	Carbonmono oxide is formed	No such.